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RESEARCH ARTICLE

A Graph-Based Heuristic for Bus Route Planning Using Taxi Origin-Destination Data

ZIHAO YU¹, (Graduate Student Member, IEEE),
BRIAN GODWIN S. LIM^{1,2,3}, (Member, IEEE), RENZO ROEL P. TAN^{1,4}, (Member, IEEE),
HIROTAKE KAJI⁵, (Member, IEEE), AND KAZUSHI IKEDA¹, (Senior Member, IEEE)

¹Division of Information Science, Nara Institute of Science and Technology, Nara 630-0192, Japan

²Graduate School of Informatics, Kyoto University, Kyoto 606-8501, Japan

³School of Science and Engineering, Ateneo de Manila University, Metro Manila 1108, Philippines

⁴John Gokongwei School of Management, Ateneo de Manila University, Metro Manila 1108, Philippines

⁵Frontier Research Center, Toyota Motor Corporation, Aichi 471-8571, Japan

Corresponding author: Zihao Yu (yu.zihao.xs9@is.naist.jp)

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ABSTRACT As the global population ages and transportation resources become increasingly constrained, mid-sized suburban aging cities face growing challenges in designing efficient and sustainable public transit systems. In response, this study introduces a comprehensive and scalable data-driven heuristic based on graph theory for bus stop selection and route planning, leveraging large-scale taxi origin-destination data. The proposed algorithm features a density-based spatial clustering module that respects pedestrian networks and physical distances to identify candidate bus stops corresponding to mobility hotspots. Subsequently, a K-medoids clustering component based on road distances further segments and assigns bus stops to bus routes while respecting road networks. Finally, a composite Hamiltonian path construction heuristic plans feasible bus routes for each segment. A case study of Susono city in Shizuoka, Japan, demonstrates the effectiveness of the proposed methodology in generating four practical, user-centric, and context-aware bus routes with high operational efficiency. Overall, the proposed algorithm provides a data-driven framework for effective and adaptive mobility planning suitable for both local authorities and bus operators.

INDEX TERMS Bus route planning, data-driven transit plan, graph theory, Hamiltonian path, K-medoids, mobility, spatial clustering, taxi origin-destination.

I. INTRODUCTION

As global populations age and urban areas undergo demographic and spatial shifts, mid-sized suburban aging cities increasingly face challenges in maintaining efficient and sustainable public transit systems under fiscal constraints [1]. In particular, the low-density settlement patterns in these regions result in diminishing ridership, which in turn increases the per-passenger operating costs, making traditional fixed-route systems unsustainable. Moreover, this degradation of public transit systems transcends mere inconvenience, evolving into a critical socioeconomic concern. Crucially, this can stifle local consumption and

disproportionately burden households that are forced into car dependency. Yet, the rising number of traffic accidents involving elderly drivers in Japan has prompted the national and local governments to encourage voluntary license surrender. The success of such policies, however, is contingent upon the availability of reliable and equitable public transportation alternatives. Notably, an inequitable access to transportation could result in social isolation for the elderly population, limiting their ability to access healthcare, engage with the community, and maintain a high quality of life [2].

To address these challenges, researchers have begun utilizing high-resolution mobility data—taxi global positioning system (GPS) trajectories, in particular—to uncover demand patterns and design adaptive transit systems [3], [4]. These origin-destination (OD) data capture authentic

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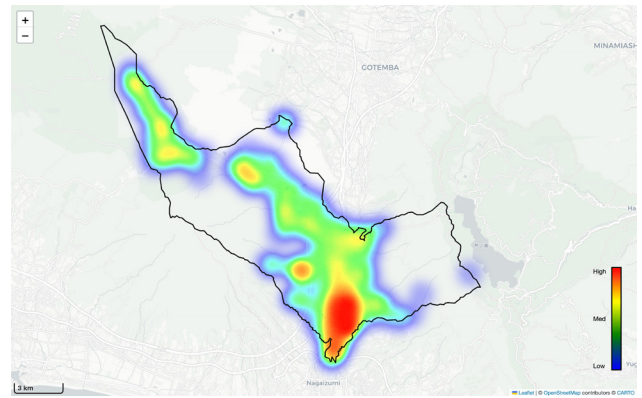
movement behavior that reflects time-varying and spatially non-uniform travel patterns [3], enabling transportation companies to detect underserved regions and dynamically adjust service coverage [5]. Crucially, a significant body of work has predominantly focused on megacities, such as Beijing [6]. However, non-metropolitan regions often exhibit different population scales, spatial structures, and travel dynamics, *e.g.*, shorter trips and high reliance on community facilities. This difference highlights the gap in existing literature in adapting and validating these data-driven planning methodologies in a different context.

Moreover, despite the advances of these techniques, many works tend to address the transit planning problem in a fragmented manner. In fact, a recent review of urban transportation network design problems highlights their complex, multi-layered nature involving strategic, tactical, and operational decisions—from bus stop placement to bus route design—that are deeply interrelated [7]. This further underscores the need for a more holistic, multi-stage framework integrating these steps to avoid suboptimal results.

Toward developing a comprehensive bus route planning methodology, the following are considered. First, density-based clustering methods are well-suited for locating OD hotspots without predefining cluster counts or assuming cluster shapes, making them ideal for identifying candidate bus stops [8]. These methods outperform traditional grid or K-means clustering in transport applications due to their robustness to noise and data irregularity [8]. Once bus stops have been identified, K-medoids clustering based on physical road distances helps segment and assign these to bus routes that respect road networks [9]. Finally, a composite heuristic algorithm for constructing Hamiltonian paths may be applied to each segment for planning feasible, intuitive, and near-optimal bus routes that minimize travel distances [10].

Susono city, located in Shizuoka Prefecture, Japan, epitomizes this planning challenge with over 30% of its population aged 65 or older. The city's terrain combines a compact urban valley with sparsely populated mountainous outskirts. This spatial heterogeneity is evident in the population density heatmap presented in Fig. 1, which shows a high concentration of residents in the southern urban corridor [11]. Moreover, the existing community bus services *Susono-ru* were discontinued in 2022 [12]. For local governments facing such demographic, spatial, and fiscal constraints, a transit design methodology that integrates actual demand behavior with infrastructure limitations offers an effective solution. In response, this study proposes a data-driven heuristic for bus stop selection and route planning, built upon three interlinked stages:

- 1) **Identify candidate bus stops.** Project taxi OD coordinates onto the pedestrian network and apply DBSCAN to identify mobility hotspots while respecting physical distances [13].
- 2) **Segment to bus routes.** Project selected bus stops onto the road network and apply K-medoids for



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FIGURE 1. Population density heatmap for Susono city.

segmenting these stops to bus routes while respecting road distances [9].

- 3) **Plan bus routes.** Construct Hamiltonian paths representing bus routes for each segment using a multi-fragment approach with a two-opt improvement procedure [10].

The proposed approach differs from purely geometric or scheduling models by integrating observed demand patterns with topological constraints, yielding transit plans that service actual mobility demands. When applied to Susono city, the proposed algorithm generated four practical, user-centric, and context-appropriate bus routes with high operational efficiency and manageable route lengths. Overall, the contributions of the paper include:

- Demonstrating the effectiveness of network-aware demand modeling and clustering for transit planning, specifically tailored to mid-sized suburban aging cities;
- Introducing a comprehensive and scalable graph-based heuristic for bus route planning that adapts to real-world road geometry and bridges the stages of stop identification, route segmentation, and route planning; and
- Providing a case study that can serve as a blueprint for other cities facing similar demographic and fiscal challenges.

The paper is organized as follows. Section II first describes the Susono dataset used. Section III then lays out the proposed graph-based heuristic. Section IV presents the results for Susono city with a particular focus on the data-driven approach, the practical considerations of the generated bus routes, and the key features of the proposed algorithm. To further demonstrate the effectiveness of the proposed algorithm, Section V provides a comparative analysis against a baseline method for generating bus routes, focusing on service coverage, bus share, and circuitry. Section VI then concludes with a summary of the work and recommendations for future studies.

II. SUSONO DATASET

This section details the primary mobility data and supplementary geographical data used in the study, as well as

the pre-processing steps performed to ensure a clean and consistent dataset for analysis.

A. TAXI ORIGIN-DESTINATION DATASET

This study used taxi OD data from Susono city.¹ The dataset consists of OD GPS coordinate logs from taxi operations between October 1, 2022, and April 9, 2024, with each entry containing the following information:

- **Spatial coordinates.** Origin (o_{lat} , o_{lon}) and destination (d_{lat} , d_{lon}) GPS coordinates;
- **Temporal information.** Timestamps for start t_{start} and end t_{end} of trips;
- **Trip distance.** Total trip distance; and
- **Identifiers.** Anonymized vehicle and driver IDs.

In total, the dataset spans over 18 months and contains tens of thousands of trip records. It also captures both weekday and weekend travel behavior across different times of day and seasons, providing sufficient granularity for analyzing spatial mobility demand patterns.

The spatial distribution of the OD coordinates is visualized as density heatmaps in Figs. 2 and 3. These figures highlight non-uniform travel activity across the city, with concentration in specific areas, reflecting the city's physical and functional geography. The southern urban basin, where the highest density is observed, serves as the city's primary hub. This area contains the main railway station (JR Susono Station), the Susono City Hall, and the majority of commercial and retail establishments. In contrast, the northern regions are more mountainous and residential, with key destinations like the Fuji Safari Park contributing to more isolated pockets of travel demand. These visualizations highlight the spatial heterogeneity of demand and underscore the need to filter for intra-city trips and focus on the high-density core for public transit planning.

Data representativeness: Taxi ridership inherently represents a specific demographic, potentially excluding a subset of the population with different mobility activities and patterns. Nevertheless, extensive literature suggests that large-scale taxi OD data sufficiently act as effective proxies for uncovering the underlying structure and functional zones of a city [3], [4], [14]. Specifically, taxi pick-up and drop-off hotspots typically correlate strongly with key locations shared by the general public, such as transport hubs, medical facilities, and commercial centers. In the context of Susono city, this alignment is visually evident in the heatmaps in Figs. 2 and 3, where taxi demand clusters around the city's primary civic and economic nodes. Hence, identifying candidate bus stops from these revealed OD hotspots ensures that the generated transit plan connects the city's most vital functional hubs, serving a utility that extends beyond the specific demographic of taxi users.

¹The dataset is from a private company and is not publicly available.

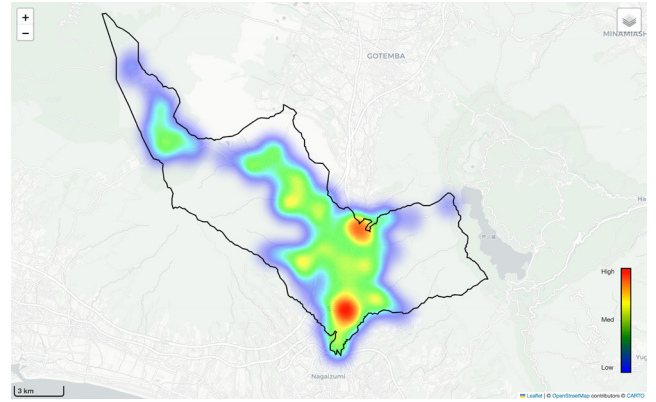


FIGURE 2. Density heatmap of taxi origin coordinates for Susono city.

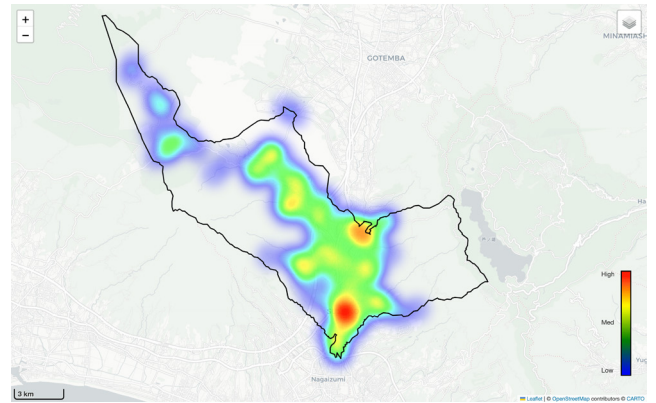


FIGURE 3. Density heatmap of taxi destination coordinates for Susono city.

B. GEOGRAPHICAL DATASETS

In addition to the above mobility dataset, two key geographical datasets were also utilized to further provide spatial context and constraints for the analysis. First, the official administrative boundary of Susono city was obtained as a GeoJSON file. This polygon served as the primary spatial filter, ensuring that only intra-city trips were considered. Second, the street network for the city was sourced from OpenStreetMap (OSM), a collaborative, open-source global map. Using the OSMnx Python package [15], the pedestrian network $\mathcal{G}_{walk}$, modeling walkways and residential streets, and the road network $\mathcal{G}_{drive}$, modeling drivable public streets and service roads, were obtained. These graphs form the topological foundation for all distance calculations and route planning tasks detailed in Section III.

C. DATA PRE-PROCESSING

To ensure consistency with the geographic scope of the study, the following pre-processing steps were performed on the raw taxi OD data:

- 1) **Spatial filter.** Only trips with both origin and destination coordinates located entirely within the official administrative boundary of Susono city were retained, removing intercity, long-distance, or rural fringe trips.

- 2) **Temporal filter.** Only trips occurring between April 2023 and April 2024 were retained, reflecting recent, post-pandemic travel patterns.
- 3) **Missing value filter.** Only trips with complete coordinate and timestamp records were retained.

Crucially, this study does not impose a fixed distance threshold, unlike prior works, *e.g.*, excluding trips shorter than 1 km or longer than 150 km. Instead, the spatial filter pre-processing step enforces strict conformance to the Susono city boundary, as shown in Figs. 2 and 3. This choice ensures that all further analyses—bus stop selection, bus route planning, and evaluation—are conducted on a geographically consistent OD dataset fully contained within the municipal area of interest for intra-city public transit analysis.

III. METHODOLOGY

This section presents the proposed six-step graph-based heuristic for bus stop selection and route planning.

A. PROJECT TAXI OD COORDINATES ONTO PEDESTRIAN NETWORK

Understanding the spatial distribution of mobility activities is essential for identifying where public transport infrastructure should be concentrated. The analysis considers over 30,000 trips, resulting in over 60,000 origin and destination coordinates. Directly clustering these raw GPS coordinates can be noisy and geographically unconstrained. Instead, these points are first projected onto a structured pedestrian network, $\mathcal{G}_{\text{walk}} = (\mathcal{V}_{\text{walk}}, \mathcal{E}_{\text{walk}})$, which better reflects the actual topology.

The projection is done by assigning each GPS point $p \in \mathcal{P}$ to its closest node in $\mathcal{V}_{\text{walk}}$

$$\text{Proj}(p; \mathcal{V}_{\text{walk}}) := \arg \min_{v \in \mathcal{V}_{\text{walk}}} d_{\text{Haversine}}(p, v), \quad (1)$$

where $d_{\text{Haversine}}$ represents the Haversine distance of two points on a sphere. After projection, each node $v \in \mathcal{V}_{\text{walk}}$ is assigned a demand weight defined as

$$w(v) := \sum_{p \in \mathcal{P}} \mathbb{1}[\text{Proj}(p; \mathcal{V}_{\text{walk}}) = v]. \quad (2)$$

This yields a scalar density signal over the set of nodes. The result of this process is an empirically grounded demand distribution overlaid on a pedestrian topology.

The choice of graph-based projection: Unlike a direct grid-based spatial aggregation, a preliminary projection of OD points onto the pedestrian network offers several advantages:

- It eliminates spurious aggregation in inaccessible areas, such as forests, lakes, or private properties;
- It preserves the connectivity and navigability of real-world pedestrian networks for subsequent analyses, ensuring equitable urban accessibility [4]; and
- It allows for seamless integration with shortest-path calculations and node-based clustering strategies.

Moreover, this approach also builds upon the idea of infrastructure-constrained demand mapping previously explored in bus stop inference and urban mobility studies [13], [15].

B. IDENTIFY CANDIDATE BUS STOPS WITH GRAPH-BASED DBSCAN

Identifying candidate bus stops requires detecting spatial concentrations of demand patterns. DBSCAN is a well-suited density-based clustering algorithm to group nodes of the pedestrian network $\mathcal{V}_{\text{walk}}$ into high-demand clusters.

DBSCAN defines clusters as sets of densely connected points described by two parameters: a radius ε and a minimum number of points $n_{\text{MinPoints}}$. A point is considered a *core point* if it has at least $n_{\text{MinPoints}}$ neighbors within its ε -neighborhood. Moreover, given the distance metric $d(u, v; \mathcal{G}_{\text{walk}})$ defining the shortest path distance between nodes u and v based on $\mathcal{G}_{\text{walk}}$, the neighborhood of a point $v \in \mathcal{V}_{\text{walk}}$ is defined as

$$\mathcal{N}_{\varepsilon}(v) := \{u \in \mathcal{V}_{\text{walk}} \mid d(u, v; \mathcal{G}_{\text{walk}}) \leq \varepsilon\}. \quad (3)$$

A cluster is then formed by all points that are reachable from core points.

The choice of graph-based distance: While traditional DBSCAN implementations use the Euclidean distance, this choice of metric inherently assumes straight-line accessibility. However, when applied to regions with irregular and non-Euclidean topologies, the results produced may become misleading, especially in areas separated by mountains, rivers, or highways. To address this, the shortest path distance over the pedestrian network $d(u, v; \mathcal{G}_{\text{walk}})$ is chosen similarly to previous works [13]. This graph-based metric respects real-world connectivity and pedestrian reachability, thereby improving clustering reliability in irregular urban spaces.

C. SELECT REPRESENTATIVE BUS STOPS VIA REDUNDANCY FILTERING

The number of candidate bus stops identified with DBSCAN often exceeds what is operationally viable. Placing bus stops too densely may lead to unnecessary detours, increased travel time, and excessive operational costs. To balance service coverage with operational efficiency, a spatial redundancy filtering step is performed to select representative bus stops within a predefined radius.

Let $\mathcal{C}_{\text{walk}} = \{c_1, c_2, \dots, c_C\}$ denote the set of candidate bus stops identified with DBSCAN in the previous step. For each $c \in \mathcal{C}_{\text{walk}}$, the neighborhood of candidate stop c within a redundancy radius r is defined as

$$\mathcal{N}_r(c) := \{c' \in \mathcal{C}_{\text{walk}} \mid d(c', c; \mathcal{G}_{\text{walk}}) \leq r\}. \quad (4)$$

From each neighborhood, only the node with the highest demand weight defined in (2) is retained, while other nodes are filtered to ensure that bus stop distances respect human-scale mobility patterns while also reducing redundancy. The selected bus stops are then denoted as $\mathcal{S}_{\text{walk}}$.

The choice of 500 m to 600 m redundancy radius: The redundancy radius r is chosen between 500 m and 600 m, as validated by both empirical transit efficiency and user satisfaction studies. Reference [16] demonstrated that although passengers often underestimate their walking distance, dissatisfaction significantly increases once the distance to a bus stop exceeds 500 m. Their GPS-validated survey revealed that “95% of respondents lived within 550 meters of the closest bus stop,” yet many self-reported this as shorter, indicating psychological resistance to longer access distances. Complementing this, [17] found that optimal stop spacing in urban environments typically falls between 400 m and 600 m, depending on ridership levels, access mode, and congestion penalties. These findings suggest that a 500 m to 600 m radius effectively balances operational efficiency and user satisfaction.

D. PROJECT BUS STOPS ONTO ROAD NETWORK

After identifying and selecting the bus stops through clustering and filtering, a critical transition step is to map these from pedestrian network $\mathcal{G}_{\text{walk}}$ onto road network $\mathcal{G}_{\text{drive}} = (\mathcal{V}_{\text{drive}}, \mathcal{E}_{\text{drive}})$. This process ensures that the subsequent bus route planning conforms to its actual operating environment and constraints. For instance, areas with high pedestrian accessibility, such as the center of parks and plazas, may be feasible OD hotspots, but buses cannot enter these locations directly. Hence, these theoretical stops must be projected onto adjacent roads where vehicles can physically stop and traverse. The effectiveness of public transit depends not only on service area coverage but also heavily on the seamless transition for passengers from the pedestrian to the vehicular environment [18].

Given the above, the selected bus stops $\mathcal{S}_{\text{walk}}$ are projected to their closest node in $\mathcal{V}_{\text{drive}}$ obtaining

$$\mathcal{S}_{\text{drive}} := \left\{ \arg \min_{v \in \mathcal{V}_{\text{drive}}} d_{\text{Haversine}}(s, v) \mid s \in \mathcal{S}_{\text{walk}} \right\}. \quad (5)$$

The choice of graph-based projection: Similar to the first step, projecting the selected bus stops onto the road network ensures that all subsequent analyses and calculations are performed on actual traversable paths for buses, thus laying the foundation for constructing feasible and efficient bus routes.

E. SEGMENT BUS ROUTE WITH K-MEDOIDS

Once the projected bus stops have been mapped onto the road network, the next step is to segment and assign these into K bus routes. The K-medoids clustering offers two primary advantages for this task:

- 1) **Interpretability of cluster centers.** K-medoids selects an actual data point within a cluster as the cluster center or medoid, providing interpretability to the cluster representative for planning and comprehension [19].
- 2) **Compatibility with arbitrary metrics.** The algorithm can also directly work with an arbitrary distance metric,

removing the reliance on coordinates and Euclidean distance.

K-medoids clustering aims to segment the data points into K partitions $\mathcal{S}_{\text{drive}} = \mathcal{S}_{\text{drive}}^{(1)} \cup \mathcal{S}_{\text{drive}}^{(2)} \cup \dots \cup \mathcal{S}_{\text{drive}}^{(K)}$ such that $\mathcal{S}_{\text{drive}}^{(i)} \cap \mathcal{S}_{\text{drive}}^{(j)} = \emptyset$ for all $i \neq j$, $1 \leq i, j \leq K$ and the sum of the distances from non-medoid points $p^{(k)} \in \mathcal{S}_{\text{drive}}^{(k)}$ to the cluster medoid $p_{\text{medoid}}^{(k)}$ is minimized across all $1 \leq k \leq K$. Similar to the DBSCAN employed in the first step, the K-medoids clustering uses the distance metric $d(u, v; \mathcal{G}_{\text{drive}})$ defining the shortest path distance between nodes u and v based on $\mathcal{G}_{\text{drive}}$ to accurately reflect physical distances required for buses. The objective function may then be expressed as

$$\min \sum_{k=1}^K \sum_{p^{(k)} \in \mathcal{S}_{\text{drive}}^{(k)}} d(p^{(k)}, p_{\text{medoid}}^{(k)}; \mathcal{G}_{\text{drive}}). \quad (6)$$

The choice of graph-based distance: To reiterate, in the context of urban transport planning, using the shortest path distance over the road network $d(u, v; \mathcal{G}_{\text{drive}})$ significantly improves the accuracy of spatial analysis as it properly accounts for real-world connectivity and topological constraints, such as rivers, buildings, or one-way streets [20].

F. HEURISTICALLY PLAN BUS ROUTES

Once the projected bus stops have been segmented and assigned to one of K bus routes, the final step is to plan the actual bus path for each of the K routes. This problem is essentially equivalent to optimizing Hamiltonian paths, which seeks to find a path that visits every bus stop in each route exactly once while minimizing the total travel distance.

Solving for the optimal Hamiltonian path is a well-known variant of the famous Traveling Salesman Problem (TSP) [21]. However, obtaining the exact solution to both problems is known to be NP-hard, making its computation prohibitively expensive even for a moderate-sized instance. A common alternative is to adopt a heuristic approach that effectively strikes a balance between computational complexity and optimality. This paper utilizes a composite heuristic algorithm comprising two stages—a multi-fragment approach for constructing Hamiltonian paths with a two-opt improvement procedure.

1) MULTI-FRAGMENT APPROACH FOR HAMILTONIAN PATH CONSTRUCTION

The multi-fragment approach for constructing a Hamiltonian path is a greedy algorithm inspired by Kruskal’s algorithm for constructing a minimum spanning tree. Concretely, for a set of nodes corresponding to a bus route, the algorithm iteratively connects the pair of *open* nodes, *i.e.*, nodes with degree less than 2, with the smallest distance $d(u, v; \mathcal{G}_{\text{drive}})$ among all current path fragments. The degree constraint for defining *open* nodes ensures that the resulting subgraph is a Hamiltonian path without a cycle. This greedy heuristic

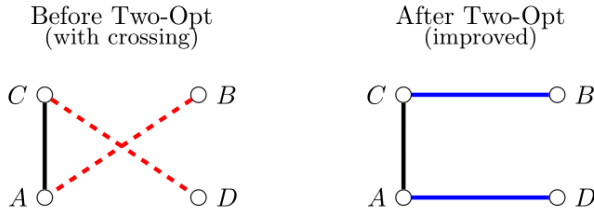


FIGURE 4. Schematic of the two-opt algorithm.

efficiently constructs a near-optimal Hamiltonian path, laying the groundwork for subsequent refinement.

2) PATH REFINEMENT WITH TWO-OPT LOCAL SEARCH

The Hamiltonian path obtained from the preceding greedy heuristic algorithm, while near-optimal, is not guaranteed to be globally optimal. To further refine the constructed path, improvement heuristics may be implemented [10]. The two-opt improvement procedure is a simple yet powerful local search technique whose core idea is to shorten a path by eliminating crossings, as illustrated in Fig. 4. In particular, the algorithm systematically examines every pair of non-adjacent edges in the path. For every such pair, it tries to eliminate path crossings by swapping two crossing edges (red dashed lines) and keeps this swap if the new path (blue solid lines) has a shorter total length. This process is repeated for the modified path after every acceptance of a swap until no further improvements can be found. The condition for this swap can be expressed as

$$d(A, B) + d(C, D) > d(A, D) + d(C, B). \quad (7)$$

Overall, by combining a greedy heuristic construction with two-opt refinement, a near-optimal Hamiltonian path traversing every bus stop for each bus route is constructed, thus completing the design of the entire bus route network.

IV. RESULTS

As an application of the proposed multi-stage, graph-based heuristic for bus stop selection and route planning, a transit network plan composed of four independent bus routes was generated for Susono city. This section provides a comprehensive presentation and analysis of the results, structured around the data-driven approach of the methodology, the practical considerations of the generated bus routes, as well as the key features of the proposed algorithm.

A. ALGORITHM RESULTS

The results for each step of the proposed methodology are presented below.

1) PROJECT TAXI OD COORDINATES ONTO PEDESTRIAN NETWORK

In the context of Susono city, the pedestrian network $\mathcal{G}_{\text{walk}}$ was obtained from OpenStreetMap through the `OSMnx` Python package [15]. The network captures minor residential

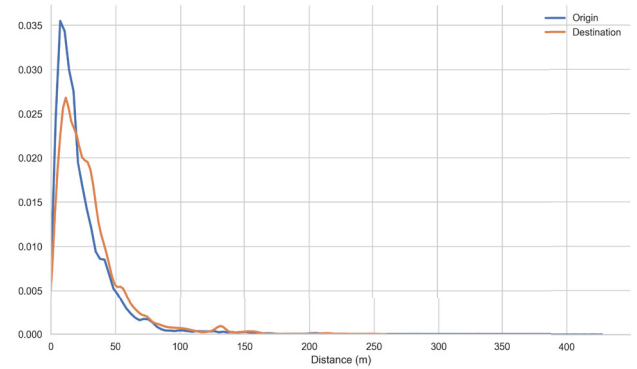


FIGURE 5. Distribution of projection distances for taxi OD coordinates onto Susono city's pedestrian network.

streets, pedestrian walkways, and legally accessible non-motorized paths. This network was carefully filtered to exclude major highways or tunnels not intended for pedestrian access. Each taxi OD coordinate was then projected onto its nearest node in $\mathcal{V}_{\text{walk}}$ using the `nearest_nodes` method. To reiterate, only trips with both OD coordinates located within the official administrative boundary of Susono city were considered to focus the analysis strictly on intra-city mobility.

The distribution of these projection distances is presented in Fig. 5. The plot reveals that the vast majority of coordinates are located in very close proximity to a node in $\mathcal{V}_{\text{walk}}$, with a strong peak below 50 meters. This validates the high resolution of the underlying street network from OpenStreetMap and confirms the accuracy of the projection process.

Figure 6 then presents the demand distribution overlaid on the pedestrian topology. In particular, blue nodes represent the projected OD points onto the pedestrian network, while red edges denote pedestrian-accessible walkways or segments. It is worth noting that dense aggregations appear along the southern urban corridor where population and commerce are concentrated, while peripheral areas, especially toward Mt. Fuji and Lake Yamanaka, exhibit sparse travel activity due to terrain and limited access. These node-level demand weights on $\mathcal{G}_{\text{walk}}$ form the foundation for clustering in the next step.

2) IDENTIFY CANDIDATE BUS STOPS WITH GRAPH-BASED DBSCAN

To identify candidate bus stops, DBSCAN was employed on the pedestrian network nodes $\mathcal{V}_{\text{walk}}$ using the demand weight in Eqn. (2) and the shortest path distance metric $d(u, v; \mathcal{G}_{\text{walk}})$. The clustering parameters were then set with the aim of balancing inclusivity and significance. In particular, the radius $\varepsilon = 50$ m approximates the typical scale of micro-accessibility in Japanese residential neighborhoods, roughly corresponding to the distance across a large intersection or a short city block. The density threshold $n_{\text{MinPoints}} = 10$ then corresponds to a moderate level of activity, capturing lower-traffic yet essential community

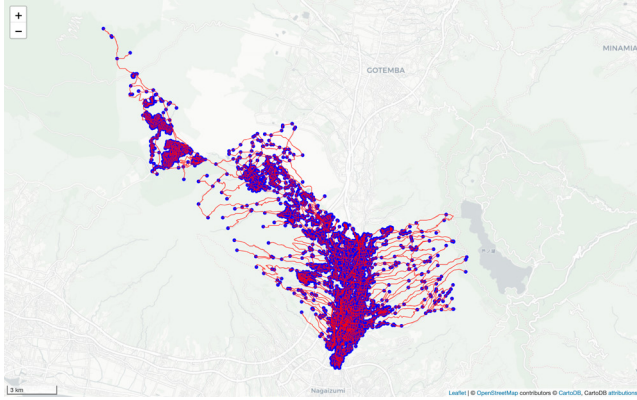


FIGURE 6. Projection of taxi OD coordinates onto Susono city's pedestrian network.

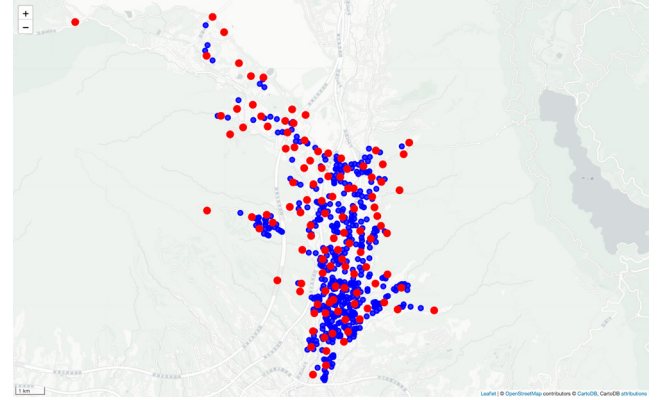


FIGURE 8. Selected bus stops for Susono city.

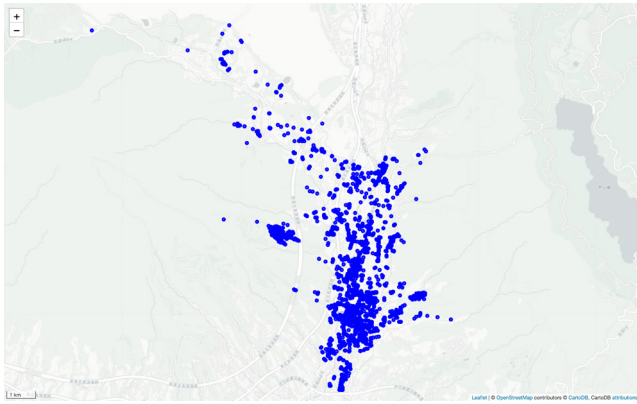


FIGURE 7. Candidate bus stops for Susono city.

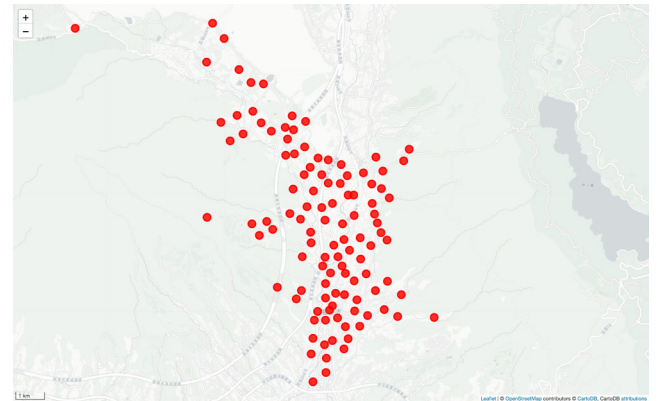


FIGURE 9. Projection of selected bus stops onto Susono city's road network.

locations, *e.g.*, small clinics or local shrines, relative to the dataset duration. These choices thus allow for the aggregation of scattered OD hotspots around diverse locations with sufficient ridership that justifies the placement of bus stops.

For each cluster identified by DBSCAN, the node with the largest demand weight was first selected as the representative candidate bus stop. A quantile-based filter retaining only the stops with a demand weight above the top 10% quantile was then employed. This two-stage approach—broad clustering followed by strict weight-based filtering—ensures that the final selection of candidate bus stops illustrated in Fig. 7 is both comprehensive in coverage and robust in demand significance.

3) SELECT REPRESENTATIVE BUS STOPS VIA REDUNDANCY FILTERING

For the context of Susono city, the redundancy radius for selecting representative bus stops was set to $r = 550\text{m}$. The bus stop candidates were then iteratively scanned, selecting the node with the largest demand weight and removing its neighboring nodes defined by $\mathcal{N}_r$. This procedure guarantees a minimum spacing between representative bus stops while preserving high-demand locations. Fig. 8 illustrates the result of this process: red points represent the selected bus stops while blue points represent bus stops removed from further

analyses. It is worth noting that the resulting bus stop distribution forms a robust yet compact basis for route planning in the next stages.

4) PROJECT BUS STOPS ONTO ROAD NETWORK

Similarly, the road network for Susono city $\mathcal{G}_{\text{drive}}$ was obtained from OSMnx. The network captures all drivable public streets as well as service roads. Each of the selected bus stops was then projected onto its nearest node in $\mathcal{V}_{\text{drive}}$ using the `nearest_nodes` method. Fig. 9 presents the resulting bus stops to be used for planning bus routes. Notably, these stops are uniformly distributed spatially, covering a significant area of Susono city.

5) SEGMENT BUS ROUTE WITH K-MEDOIDS

The number of bus routes K serves as a critical parameter in the current framework that directly influences the structure and operational feasibility of the transit plan. To ensure a robust selection process, a comprehensive evaluation for K ranging from 2 to 5 was conducted. The key metrics are presented in Table 1. The results reveal that although the average silhouette score is optimal at $K = 2$ (0.428), relying solely on this geometric metric is insufficient for practical transit planning. A partition of $K = 2$ merges functionally distinct urban zones into two oversized, monolithic clusters.

TABLE 1. Cluster summary for varying K .

K	Cluster Sizes	Std. of Cluster Sizes	Avg. Silhouette
2	{48, 63}	7.50	0.428
3	{23, 39, 49}	10.71	0.400
4	{19, 22, 31, 39}	7.85	0.352
5	{7, 19, 23, 25, 37}	9.68	0.331

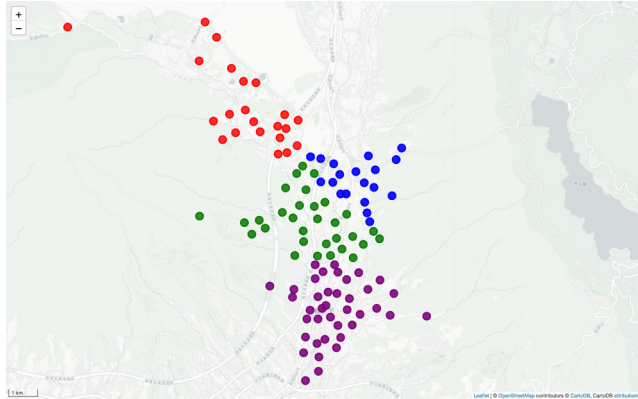


FIGURE 10. Bus route segmentation for Susono city.

From a line planning perspective, such a configuration is undesirable as it results in excessively long, circuitous routes that degrade service reliability and increase vehicle cycle times [22]. In contrast, $K = 4$ offers the optimal trade-off between statistical cohesion and operational balance. While its silhouette score (0.352) is lower than that of $K = 2$, it maintains the lowest standard deviation in route sizes (7.85) among the choices. This low standard deviation implies a more equitable distribution of stops across routes, ensuring that no single route is operationally overburdened while maintaining manageable route lengths.

Moreover, the $K = 4$ configuration naturally reveals a balanced hierarchical network structure, comprising two major backbone routes (approximately 35 stops) and two community feeder routes (approximately 21 stops). This configuration aligns with established transit network design principles that advocate for a balance between high-capacity corridors and local coverage lines to maximize system efficiency [22]. Conversely, $K = 3$ provides highly imbalanced cluster sizes, while $K = 5$ results in fragmented routes with low operational value, as indicated by the high standard deviation of 9.68. Overall, $K = 4$ provides the most operationally optimal and logically sound solution for the specific context of Susono city, effectively balancing the conflicting objectives of minimizing operational costs and maximizing service coverage. The resulting route segmentation is illustrated in Fig. 10.

Crucially, this selection process—evaluating the trade-off between geometric clustering quality and operational balance—should be repeated to determine the appropriate K when applying this framework to other contexts with potentially different mobility patterns and needs.

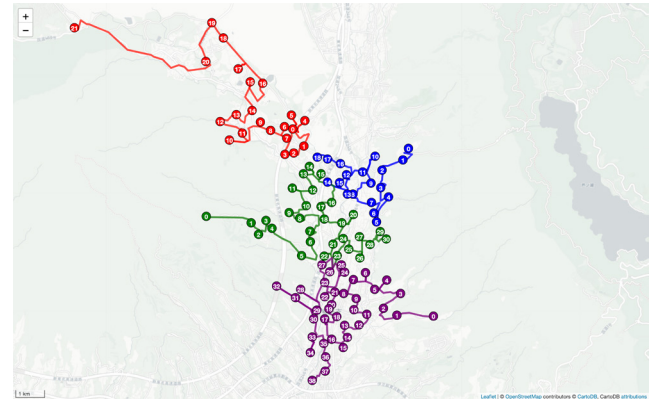


FIGURE 11. Generated bus routes for Susono city.

TABLE 2. Summary of operational metrics for the generated bus routes for Susono city.

Route Color	No. of Stops	Total Trip Distance (km)	Avg. Distance between Stops (m)
Red	22	28.5	1,359
Blue	19	15.4	857
Green	31	26.7	891
Purple	39	33.3	875

6) HEURISTICALLY PLAN BUS ROUTES

Finally, after running the composite heuristic for constructing Hamiltonian paths for each bus route, the generated transit plan consists of four functionally complementary routes as presented in Fig. 11. The four routes (Red, Blue, Green, and Purple) jointly form a service network covering the city's main areas. To quantitatively evaluate this network, key operational metrics were calculated for each route as summarized in Table 2.

Overall, the generated transit network is lean with just four routes and exhibits a hybrid functional structure of *backbone + feeder/commuter* routes. The Green and Purple routes serve as two high-capacity backbone lines covering the most densely populated areas, while the Red and Blue routes function as a core circulator and a peripheral connector, respectively, forming a network with clear functional hierarchies.

B. BUS ROUTE ANALYSES

To better understand the function served by each bus route, Fig. 12 presents the individual views of the four generated routes.

- **Red route: Northwest corridor connector.** As shown in Fig. 12, this route functions as a quintessential corridor connector. Its long, linear path spans from the north-central industrial and research area to the northwestern foothills. Notably, the route's most distant stop is strategically placed to serve the Fuji Safari Park, a major regional tourist destination. The inclusion of this

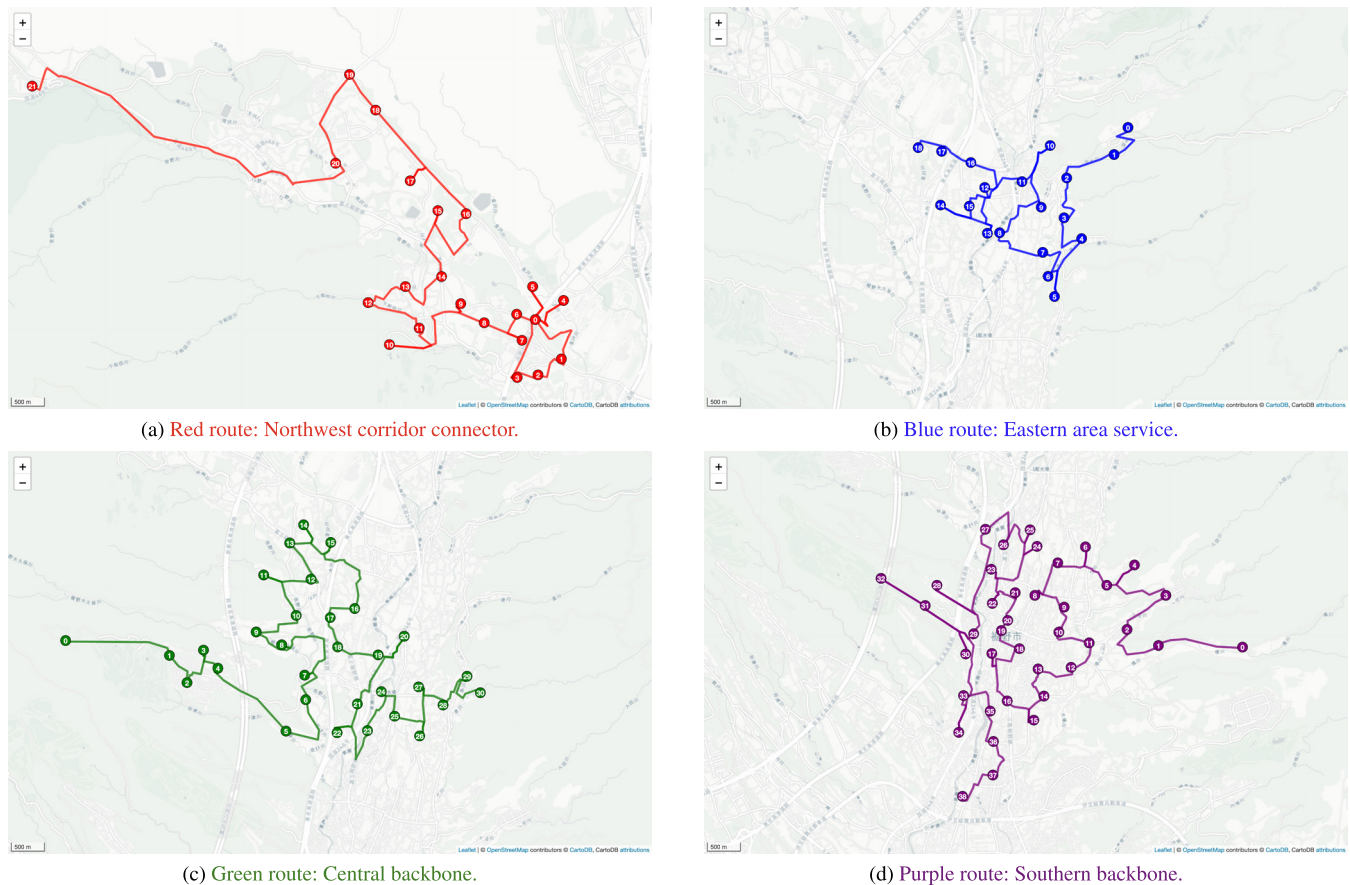


FIGURE 12. Individual views of the generated bus routes for Susono city.

spatially isolated stop is a direct result of the algorithm optimizing for service coverage. Hence, despite its distance from the city center, the graph-based DBSCAN algorithm successfully detected a significant level of activity at this location, with a total demand weight exceeding the selection threshold. Consequently, the routing heuristic extended the Red route to include this stop. Moreover, with an average distance between stops of 1,359 m—the longest among the four—this route demonstrates the algorithm’s capability to adapt to heterogeneous demand patterns, creating a vital link that serves both local industrial commuters and regional tourists.

- **Blue route: Eastern area service.** This route, illustrated in Fig. 12b, operates as a dedicated service for the city’s eastern flank. As the shortest bus route at 15.4 km, it forms a broad C-shape to service the major residential neighborhoods from the northeast, *e.g.*, Mishuku area, to the southeast. Crucially, the route operates entirely independently of the centrally located Susono Station, instead providing vital intra-regional connectivity and serving local points of interest (POIs), such as community centers and schools within the eastern districts.

- **Green route: Central backbone.** As the central backbone of the network, the Green route in Fig. 12c forms the primary north-south artery. With 31 stops, its service path is strategically aligned with the city’s main commercial streets and passes in close proximity to the most critical civic hubs, including Susono Station and the Susono City Hall area. This route is the pillar of the entire system, designed to provide high-frequency, high-capacity service connecting the most populous and functionally diverse urban corridor.
- **Purple route: Southern backbone.** The Purple route, as presented in Fig. 12d, serves as the network’s other foundational line—the southern backbone. It is the longest route at 33.3 km and has the highest number of stops at 39, providing the most extensive coverage. It provides comprehensive east-west service across the city’s vast southern built-up areas, linking numerous large residential communities with key industrial parks, which constitute major centers of employment. It functions to integrate these southern districts into the city-wide network, providing them with access to the central hubs served by the Green route.

Overall, the four generated bus routes mirror Susono’s core-periphery urban spatial structure. The two backbone

routes ensure the traffic capacity of the main north-south axis, the core circulator handles short-distance transfers within the central district, and the corridor connector solves the commuting problem for peripheral communities. These results demonstrate how the proposed algorithm can effectively translate the city's spatial language into a transit plan in a data-driven manner that is highly congruent with its morphology and function.

Nevertheless, it is worth noting that while these generated routes provide a strong data-driven foundation, they function as a preliminary transit plan. Final implementation requires further refinement, together with local authorities and the bus operator, to address operational feasibility, regulatory compliance, and limitations of the proposed graph-based heuristic.

C. KEY FEATURES OF THE GRAPH-BASED HEURISTIC

To recapitulate, this paper proposes an end-to-end graph-based bus route planning heuristic utilizing large-scale taxi OD data. When applied to the specific context of Susono city, the results reveal four intuitive and interpretable bus routes. An in-depth discussion of the algorithm's key features is presented below.

1) REVEALED POINTS OF INTEREST

While traditional transit network planning utilizes predefined POIs as the key basis for identifying bus stops [14], the proposed algorithm takes on an inverse approach: the actual travel behavior reflected in large-scale taxi OD data implicitly reveals the city's true functional hubs. The dense passenger flow in these locations underscores their importance as transportation nodes.

Crucially, the proposed algorithm aggregates both origin and destination coordinates from the taxi data to calculate the demand weights in Eqn. (2). This aggregation is a common assumption adopted in literature [14], [22] and reflects the bidirectional nature of public transit services, *e.g.*, a location serving as an origin in the morning often functions as a destination in the evening. By aggregating OD coordinates, the identified candidate bus stops capture the total potential activity of a location regardless of the time of day. The proposed algorithm thus inherently ensures that the generated bus routes directly serve the city's holistic mobility demand, thereby maximizing its socioeconomic service value.

2) EQUITABLE MOBILITY

A good transit network must not only serve hotspots, but also provide basic mobility for the general population. To evaluate this, a service coverage analysis is conducted using a synthetic population dataset for Susono city [11]. Using a 500 m (walking) buffer radius around each of the selected bus stops, the generated bus routes are able to service over 93% of the total population. Fig. 13 illustrates this, where red points represent bus stops and blue shaded areas represent serviced areas following the 500 m buffer radius. Overall, the results indicate that the proposed data-driven

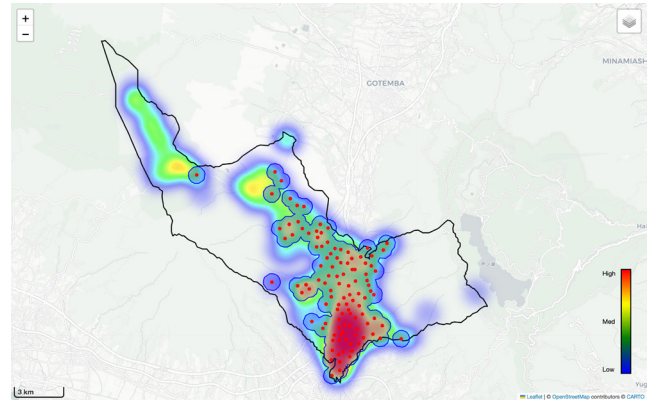


FIGURE 13. Service coverage analysis with Susono city's population.

bus routes, while only explicitly considering taxi OD data, also exhibit a high coverage of residential areas, balancing efficiency and equity.

3) PRACTICAL OPERATIONAL IMPLEMENTATION

The proposed graph-based heuristic provides a grounded topological skeleton for the transit plan. Nevertheless, bridging the gap between this theoretical model and real-world operations requires several practical considerations. The following planning decisions are identified and best discussed with the actual bus operators due to their specific operational and fiscal constraints, following previous works [14], [23].

a: ROUTE CONNECTIVITY AND TRANSFERS

The K-medoids algorithm partitions the selected bus stops into disjoint clusters to define the K bus routes. However, the physical geometries of the generated bus routes naturally exhibit significant overlaps along key corridors. For instance, the Green and Purple routes converge along the city's central north-south artery, and the Blue route physically intersects with the Green route near the central station, as shown in Fig. 11. In practical operations, these geometric overlaps may be formalized as shared transfer stops, allowing for seamless transfers between routes without requiring the clustering algorithm to explicitly model multi-route bus stops. Nevertheless, with the current configuration, the maximum walk distance for a transfer between routes is moderate, with 686 m between the Red and Green route, 585 m between the Red and Blue route, 349 m between the Blue and Green route, and 295 m between the Purple and Green route.

b: STOP PLACEMENT

The proposed algorithm projects OD hotspots onto the nearest nodes of the road network $\mathcal{G}_{\text{drive}}$ to ensure connectivity. This approach simplifies the methodology and aligns with the assumptions of previous works [14], [22]. The bus stops identified by the algorithm thus reflect strategic and potential stop locations. The actual bus stop placement may require micro-geographic considerations, such as determining the safe side of the road for boarding relative to traffic flow or

installing crosswalks. Their final curbside placement would be determined through on-site validation and collaboration with local traffic authorities to ensure pedestrian safety.

c: TEMPORAL DYNAMICS

This study utilized data aggregated over several months to design a robust, static route structure suitable for long-term infrastructure planning. While travel demand naturally fluctuates between peak/off-peak hours and weekdays/weekends, these temporal variations are best addressed through operational scheduling—specifically by adjusting headways and timetables, both of which are best left at the discretion of the bus operators [22]—rather than by altering the physical route topology. The proposed bus routes thus serve as the foundational transit plan upon which dynamic operational schedules can be optimized.

4) RELATED WORKS

Related works, such as [6], [14], [22], [23], and [24], largely focus on transit network design for metropolitan cities requiring more complex transit plans to cover diverse travel demands. Moreover, these methods also largely assume a grid-like, Euclidean topology and work with a predefined set of candidate bus stops. In contrast, the context of the current work on mid-sized suburban aging cities requires a much simpler transit plan accounting for irregular, non-Euclidean topology and identifying candidate bus stops in a data-driven manner to achieve a high level of service coverage. This offers a crucial insight for transit planning in this context: rather than aiming for network complexity and breadth, the focus should shift towards identifying and connecting core demand nodes through observed data.

The proposed multi-stage, graph-based heuristic algorithm draws inspiration from foundational methods in transit network design. The concept of decomposing the complex route planning problem into sequential sub-problems—first identifying bus stops, followed by route generation—aligns with the classic two-phase framework proposed in [23]. While their work focused on frequency optimization for a given route skeleton, the proposed algorithm adapts this sequential logic to the context of data-driven demand, ensuring that the topological structure of the network is established before detailed bus route planning.

In fact, the strategy of decomposing a large-scale routing problem into manageable components mirrors recent advancements in optimization, such as the Bi-objective Routing Decomposition (BiRD) algorithm for school bus routing [24]. This work demonstrates that breaking down the global network design problem into stop placement, routing, and scheduling sub-problems allows for the application of specialized algorithms at each stage. The proposed algorithm reflects this principle by leveraging specific graph-based heuristics for each phase of the planning process, validating the robustness of a decomposition-based strategy for modern, data-rich transit planning.

Moreover, in contrast to other works which often consider bus stop identification, route design and optimization as separate, isolated problems, the proposed framework addresses the issue raised by [7] and takes on a more holistic, comprehensive, and multi-staged approach where the representative bus stops using DBSCAN and redundancy filtering directly becomes the input for the K-medoids bus route segmentation, whose output, in turn, seamlessly feeds into the heuristic bus route planning algorithm. This pipeline ensures consistency across decision-making stages and is a key methodological feature of the present work.

Furthermore, the construction of the proposed bus routes, particularly the segmentation into distinct routes via K-medoids, is supported by the line planning models reviewed in [22]. Their comprehensive survey highlights the effectiveness of hierarchical network structures in balancing the conflicting objectives of minimizing operational costs and maximizing passenger service quality. The selection of $K = 4$ in the current paper to create a hybrid backbone-and-feeder system directly reflects these principles, ensuring that high-demand corridors are served efficiently while maintaining coverage for peripheral regions.

Overall, the proposed algorithm comprehensively covers the key stages of transit network design identified in previous works while also aligning with their algorithmic design principles. Its key features lie with its multi-step framework comprehensively covering the various planning stages, graph-based approach accounting for non-Euclidean topology, and service coverage priority for reaching peripheral regions. These result in the generation of practical, user-centric, and context-aware bus routes tailored for suburban contexts, such as Susono city.

V. COMPARATIVE ANALYSIS

To further demonstrate the effectiveness of the proposed graph-based heuristic algorithm in generating feasible bus routes for Susono city, a comparative analysis with an existing algorithm is conducted.

A. B-PLANNER BASELINE ALGORITHM

The B-Planner algorithm is a two-phase framework for planning bidirectional night bus routes [14]. Unlike most baseline algorithms, which rely on pre-defined stop locations [22], [23], [24], it also leverages large-scale taxi OD data, similar to the proposed algorithm. In the first phase, taxi OD coordinates are first discretized into grid cells to identify *hot* areas with dense passenger activity. Large *hot* areas are subsequently split into more compact clusters before selecting a candidate bus stop from each cluster. In the second phase, a bus route graph is first constructed using the candidate bus stops, bus operation constraints, and predetermined origin and destination coordinates for the bus route. Heuristic rules are then used to prune infeasible bus stops and edges iteratively based on constraints such as maximum total travel time and operation frequency. A bidirectional probability-based spreading (PBS) algorithm is then employed on the

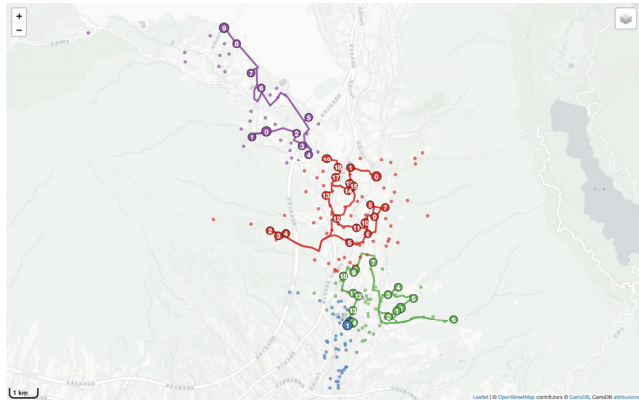


FIGURE 14. Generated bus routes for Susono city using a modified B-Planner.

constructed graph to generate candidate bus routes before selecting the optimal bidirectional route that maximizes the number of passengers under the constraints listed in Section III of [14].

The data-driven approach employed by B-Planner in leveraging taxi OD data makes it highly comparable to the proposed graph-based heuristic algorithm in generating potential bus routes for Susono city. Nevertheless, the baseline algorithm is designed to optimize only a *single* high-demand corridor based on a predetermined bus route origin and destination coordinates. Hence, this paper considers a modification of the original baseline algorithm, utilizing K-medoids clustering for generating multiple bus routes and a flow-based heuristic for identifying bus route OD coordinates within each cluster. The generated bus routes using this modified B-Planner algorithm are presented in Fig. 14, where unnumbered nodes represent candidate bus stops that fail to satisfy the constraints imposed and thus are not included in the routes. Even with the modifications, the algorithm still implicitly assumes urban terrains with grid-like Euclidean topology. In contrast, the graph-based heuristic proposed in the current study generates *multiple* bus routes while also operating on the pedestrian and road networks with potentially non-Euclidean topology, thereby directly accounting for accessibility of the proposed bus stops. These key features make the proposed algorithm more flexible and adaptive towards suburban Susono with more complex behavioral and topological patterns compared to metropolitan cities.

To objectively evaluate the utility of the two algorithms—a modified B-Planner baseline algorithm and the proposed graph-based heuristic algorithm—in generating bus routes that effectively serve the mobility demand of Susono city, an analysis based on OpenTripPlanner (OTP) was conducted. Specifically, OTP was primarily used as an independent and objective multi-modal routing engine. In particular, for every taxi OD data, OTP generated feasible itineraries, combining both walking and bus riding modes, using the bus routes generated by the two algorithms independently under the same routing parameters. Crucially, OTP provides

a way for objective service coverage, bus share, and circuitry analyses between the two generated bus routes. This approach separates the bus route planning stage from the evaluation stage, enabling fair and context-specific analyses tailored to Susono's suburban topology.

It is also worth noting that while time-based metrics may be estimated in the context of metropolitan areas characterized by grid-like, Euclidean topologies with predictable traffic flows [25], this procedure would introduce larger variance and errors in the context of mid-sized suburban aging with irregular, non-Euclidean topologies, such as the mountainous terrain of Susono city. Moreover, these metrics would also necessitate scheduling the bus routes, which requires considering the specific constraints of the bus operator [23]. Nevertheless, the fundamental transportation theory posits a strong positive correlation between distance and travel time in non-congested networks [26]. Hence, the distance-based metrics presented below represent an immutable constraint imposed by the city's actual topology, ensuring a more robust analysis and a more stable foundation for long-term infrastructure planning in irregular geographies.

B. SERVICE COVERAGE ANALYSIS

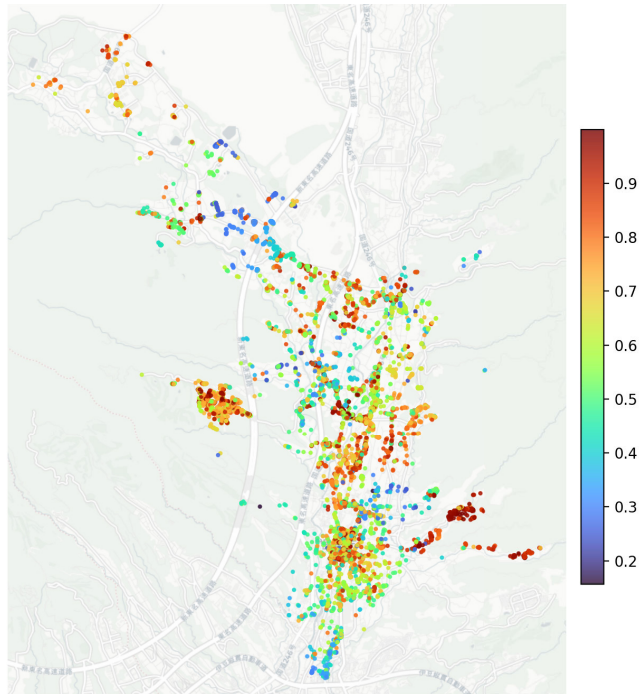
Service coverage is one of the most fundamental performance indicators for public transport systems, especially in low-density or suburban contexts. Prior works in rural and semi-urban mobility largely agree that in areas with dispersed land-use patterns, limited service frequency, and car-dependent travel behavior, the availability of *any* workable public transit option often has higher social and accessibility value than optimizing route directness or minimizing travel time [2], [18]. In particular, service coverage exerts a stronger socio-economic influence than purely geometric or efficiency-based metrics [2], [27]. In fact, for these regions, increasing the number of residents who can complete end-to-end journeys using transit is frequently regarded as the primary objective of network design, as it directly affects accessibility for elderly populations, students, and non-driving households [27]. For the current analysis, service coverage was based on the number of taxi OD data with the OTP itinerary using at least one of the bus routes.

The results in Table 3 show that the bus routes generated by the B-Planner baseline algorithm provide a moderate level of coverage, serving 67.9% (27,790 out of 40,917) of the total taxi OD data. In contrast, the bus routes generated by the proposed graph-based heuristic algorithm serve 85.9% (35,145 out of 40,917) of the total OD data, representing an 18% improvement in service coverage compared to the baseline. Moreover, the bus routes generated by the proposed algorithm also uniquely serve 9,306 OD data that the baseline does not, while only 1,951 OD data are served exclusively by the baseline algorithm.

In the specific context of Susono city, even modest improvements in service coverage are practically significant. A slightly longer or more circuitous trip remains far

TABLE 3. Service coverage analysis for Susono city.

Metric	Count	Rate (%)
Total OD data	40,917	100.0
Served by B-Planner	27,790	67.9
Served by graph-based heuristic	35,145	85.9
Served by <i>both</i>	25,839	63.2
Served by graph-based heuristic only	9,306	22.7
Served by B-Planner only	1,951	4.8
Not served by both	3,821	9.3

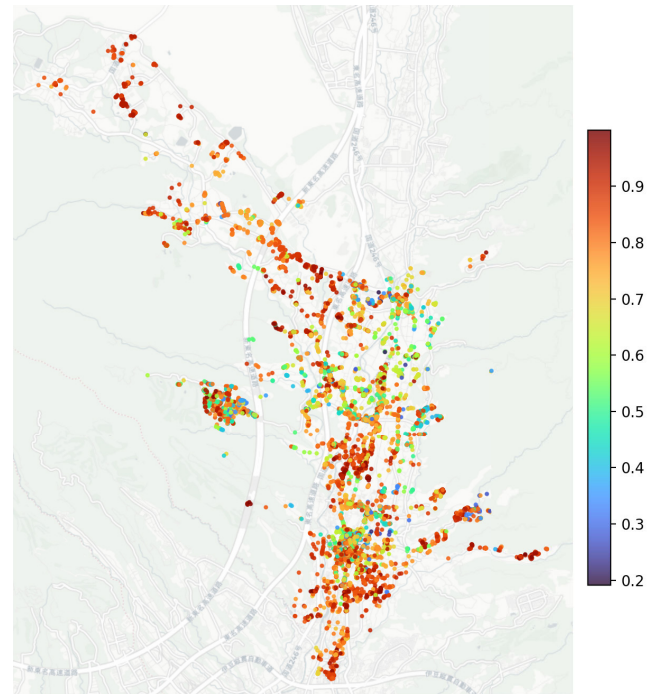
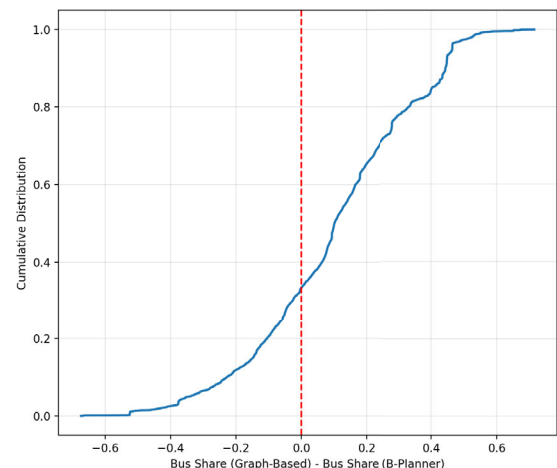
**FIGURE 15.** Bus share heatmap using B-Planner for Susono city.

more valuable than a trip that cannot be completed by available public transit. The bus routes generated by the proposed graph-based heuristic algorithm, therefore, align more closely with Susono's socio-spatial mobility needs than the baseline.

C. BUS SHARE ANALYSIS

Figures 15 and 16 present the heatmaps of bus share—the ratio of the total bus distance to the total trip (bus and walk) distance for each itinerary—across all taxi data by the origin coordinates using the bus routes generated by the B-Planner baseline and the proposed graph-based heuristic, respectively. Across nearly all spatial clusters, the proposed graph-based heuristic produces substantially higher bus share metrics, indicating a greater proportion of the OTP itineraries relying on the proposed bus routes.

To further quantify this result, Fig. 17 presents the cumulative distribution of the pairwise difference in bus share for each taxi OD data served by both the proposed bus routes and the baseline bus routes. The figure shows that in

**FIGURE 16.** Bus share heatmap using graph-based heuristic for Susono city.**FIGURE 17.** Cumulative distribution of pairwise difference in bus share for Susono city.

approximately 68% of the taxi OD data, the proposed bus routes result in higher bus share with an average increase of 14% in the metric. The implication is operationally meaningful: a higher bus share reflects stronger utilization of public transit systems, greater justification for fixed-route operational subsidies, and increased behavioral adoption of bus travel. In suburban environments such as Susono city, where transit visibility and habitual usage are key determinants of long-term ridership growth, this increase in bus usage represents a substantive improvement over the baseline.

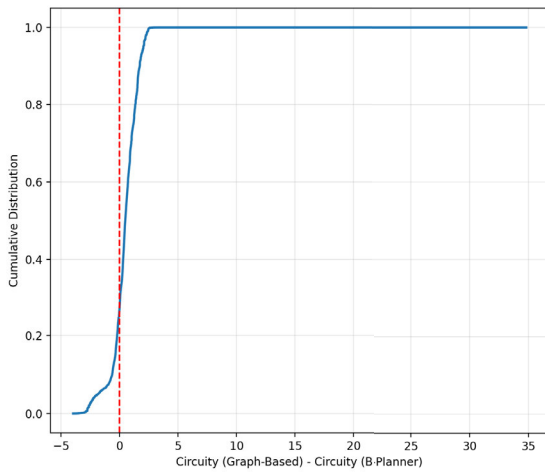


FIGURE 18. Cumulative distribution of pairwise difference in circuitry for Susono city.

D. CIRCUITY ANALYSIS

Although the proposed bus routes improve service coverage and bus share, they also increase circuitry—the ratio of the total trip distance to the total taxi trip distance in the OD data—as shown in Figs. 18 and 19. Several structural and behavioral factors explain this effect:

- 1) **Tolerance for moderately longer trips.** The OTP planning engine prioritizes simpler itineraries over total trip distance.
- 2) **Rural, non-Euclidean road networks induce higher circuitry.** Susono's irregular road networks inherently force detours, especially with the proposed graph-based heuristic prioritizing accessibility for peripheral areas.
- 3) **Difference in bus route planning objectives.** B-Planner directly optimizes high-demand corridors under operational constraints, while the proposed graph-based heuristic optimizes service coverage.

Hence, higher circuitry is expected given the behavioral and coverage priorities embedded in the two algorithms. Nevertheless, the number of legs or segments in the OTP itineraries is roughly the same for both alternatives, as shown in Fig. 20. Given the specific context of Susono city, better service coverage and bus share translate to greater practical value over better circuitry.

E. RUNTIME ANALYSIS

To further assess the practical utility and feasibility of the proposed algorithm, a runtime analysis was conducted comparing the graph-based heuristic against the B-Planner baseline. The experiments were performed on a workstation with an Apple M1 Max CPU and 32 GB of RAM.

As shown in Table 4, the proposed algorithm required a total runtime of approximately 230 seconds to process the dataset containing over 30,000 OD data and generate the bus routes, whereas the B-Planner baseline required approximately 390 seconds. Notably, a critical distinction

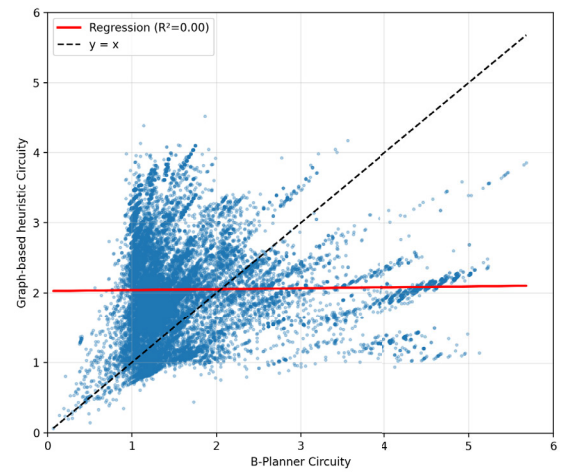


FIGURE 19. Scatterplot of circuitry for Susono city.

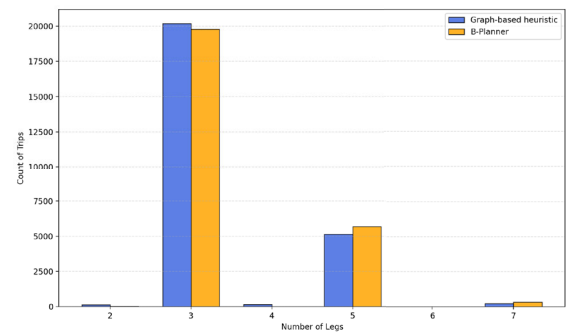


FIGURE 20. Comparative histogram of the number of legs for Susono city.

lies in the distribution of computational load. For the proposed heuristic, 96.9% of the total runtime (221.60 s) is dedicated to the pre-processing step of all-pairs shortest path calculation using Dijkstra's algorithm for both $\mathcal{G}_{\text{walk}}$ and $\mathcal{G}_{\text{drive}}$. Crucially, this is a one-time calculation that depends solely on the network topologies and not the OD dataset. The core algorithm—spanning graph-based DBSCAN, K-medoids clustering, and Hamiltonian path generation—is extremely lightweight, executing in approximately 7 seconds. In contrast, the B-Planner baseline allocates the majority of its runtime (79.4% or 312.58 s) to calculating the global Flow Matrix (FM) and Time Matrix (TM) between all candidate stops. This data aggregation step involves computationally expensive calculations that grow with the OD dataset. Overall, the analysis demonstrates that the proposed algorithm is highly scalable. Once the static network graphs are processed, it can rapidly iterate on changing demand data or parameter adjustments (*e.g.*, varying K or ϵ) in near real-time, making it superior for interactive transit planning.

F. OVERALL ANALYSIS

The analyses above underscore how the proposed graph-based heuristic efficiently generated bus routes that are more ideal

TABLE 4. Runtime analysis between the graph-based heuristic and B-Planner.

Algorithm	Time (s)	Share (%)
Graph-Based Heuristic		
Pre-processing	221.60	96.9
Core Algorithm	7.19	3.1
Total Runtime	228.79	100.0
B-Planner		
Stop Identification	1.39	0.4
FM & TM Calculation	312.58	79.4
Clustering & PBS	79.68	20.2
Total Runtime	393.65	100.0

for the specific context of Susono city compared to the baseline bus routes generated by B-Planner. Through the OTP-based comparative analysis, the proposed bus routes are shown to achieve:

- **Higher service coverage** which extends mobility to far more residents; and
- **Higher bus share** which indicates stronger utilization of fixed-route transit.

Nevertheless, given the irregular road network topology and the objective of addressing the mobility demands of Susono city, a trade-off in terms of increased circuitry is unavoidable. Overall, by directly considering the non-Euclidean topology with a graph-based approach and prioritizing service coverage for greater mobility, the bus routes generated are more practical, user-centered, and context-appropriate.

VI. CONCLUSION

In summary, this work proposes a comprehensive and scalable graph-based heuristic for bus stop selection and route planning, utilizing large-scale taxi OD data, specifically designed for mid-sized suburban aging cities. In particular, it integrates a graph-based DBSCAN and a redundancy filter to first select representative bus stops, a K-medoids clustering to then segment and assign bus stops to bus routes, and a composite heuristic algorithm for constructing Hamiltonian paths to finally plan each bus route. When applied to Susono city, the proposed algorithm generated a thin transit network plan consisting of four intuitive bus routes that effectively address the mobility needs of the city. This provides the authorities with a preliminary data-driven transit plan that may be used for further discussions and refinements with the community and bus operators before actual implementation.

Future works may consider incorporating interventions and manual adjustments from planners across each stage of the proposed framework to make it more flexible and robust. This may then be subsequently applied to other small to mid-sized suburban aging cities with potentially different needs. The proposed algorithm may also be evaluated on regions where both taxi OD data and public transit systems are available to

comprehensively evaluate the effectiveness of the generated bus routes as a potential alternative.

REFERENCES

- [1] C. Daraio, M. Diana, F. Di Costa, C. Leporelli, G. Matteucci, and A. Nastasi, "Efficiency and effectiveness in the urban public transport sector: A critical review with directions for future research," *Eur. J. Oper. Res.*, vol. 248, no. 1, pp. 1–20, Jan. 2016.
- [2] I. Shergold and G. Parkhurst, "Transport-related social exclusion amongst older people in rural southwest England and Wales," *J. Rural Stud.*, vol. 28, no. 4, pp. 412–421, Oct. 2012.
- [3] J. Tang, F. Liu, Y. Wang, and H. Wang, "Uncovering urban human mobility from large scale taxi GPS data," *Phys. A, Stat. Mech. Appl.*, vol. 438, pp. 140–153, Nov. 2015.
- [4] X. Liu, L. Gong, Y. Gong, and Y. Liu, "Revealing travel patterns and city structure with taxi trip data," *J. Transp. Geography*, vol. 43, pp. 78–90, Feb. 2015.
- [5] S. P. Chuah, H. Wu, Y. Lu, L. Yu, and S. Bressan, "Bus routes design and optimization via taxi data analytics," in *Proc. 25th ACM Int. Conf. Inf. Knowl. Manage.*, Oct. 2016, pp. 2417–2420.
- [6] J. Yuan, Y. Zheng, C. Zhang, W. Xie, X. Xie, G. Sun, and Y. Huang, "T-drive: Driving directions based on taxi trajectories," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geographic Inf. Syst.*, Nov. 2010, pp. 99–108.
- [7] R. Z. Farahani, E. Miandoabchi, W. Y. Szeto, and H. Rashidi, "A review of urban transportation network design problems," *Eur. J. Oper. Res.*, vol. 229, no. 2, pp. 281–302, Sep. 2013.
- [8] M. Ester, H. Kriegl, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in *Proc. 2nd Int. Conf. Knowl. Discovery Data Mining*, 1996, pp. 226–231.
- [9] H.-S. Park and C.-H. Jun, "A simple and fast algorithm for K-medoids clustering," *Expert Syst. Appl.*, vol. 36, no. 2, pp. 3336–3341, Mar. 2009.
- [10] G. A. Croes, "A method for solving traveling-salesman problems," *Oper. Res.*, vol. 6, no. 6, pp. 791–812, Dec. 1958.
- [11] T. Murata, T. Harada, and D. Masui, "Comparing transition procedures in modified simulated-annealing-based synthetic reconstruction method without samples," *SICE J. Control, Meas., Syst. Integr.*, vol. 10, no. 6, pp. 513–519, Nov. 2017.
- [12] (2022). *Susono-RU Has Been Discontinued | Susono City*. Accessed: Jul. 25, 2025. [Online]. Available: <https://www.city.susono.shizuoka.jp/kurashi/4/1/9165.html>
- [13] W. Wang, L. Tao, C. Gao, B. Wang, H. Yang, and Z. Zhang, "A C-DBSCAN algorithm for determining bus-stop locations based on taxi GPS data," in *Proc. Int. Conf. Adv. Data Mining Appl.*, 2014, pp. 293–304.
- [14] C. Chen, D. Zhang, N. Li, and Z.-H. Zhou, "B-planner: Planning bidirectional night bus routes using large-scale taxi GPS traces," *IEEE Trans. Intell. Transp. Syst.*, vol. 15, no. 4, pp. 1451–1465, Aug. 2014.
- [15] G. Boeing, "OSMnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks," *Comput., Environ. Urban Syst.*, vol. 65, pp. 126–139, Sep. 2017.
- [16] F. Balducci, A. Ståhl, O. Svensson, B. Jonsson, Y. Westerlund, J. Dolcini, and C. Chiatti, "How far is the closest bus stop? An evaluation of self-reported versus GIS-computed distance to the bus among older people and factors influencing their perception of distance," *Geomatics*, vol. 3, no. 4, pp. 580–596, Dec. 2023.
- [17] A. Tirachini, "The economics and engineering of bus stops: Spacing, design and congestion," *Transp. Res. A, Policy Pract.*, vol. 59, pp. 37–57, Jan. 2014.
- [18] A. El-Geneidy and D. Levinson, "Mapping accessibility over time," *J. Maps*, pp. 76–87, 2007.
- [19] L. Kaufman and P. J. Rousseeuw, *Finding Groups in Data: An Introduction to Cluster Analysis*. Hoboken, NJ, USA: Wiley, 2009.
- [20] A. Okabe, K.-I. Okunuki, and S. Shiode, "SANET: A toolbox for spatial analysis on a network," *Geographical Anal.*, vol. 38, no. 1, pp. 57–66, Jan. 2006.
- [21] E. L. Lawler, J. K. Lenstra, A. H. G. R. Kan, and D. B. Shmoys, "The traveling salesman problem: A case study," in *The Traveling Salesman Problem: A Guided Tour of Combinatorial Optimization* (Wiley-Interscience Series in Discrete Mathematics), vol. 12, E. L. Lawler, J. K. Lenstra, A. H. G. R. Kan, and D. B. Shmoys, Eds., Chichester, U.K.: Wiley, Sep. 1985, pp. 1–12.
- [22] A. Schöbel, "Line planning in public transportation: Models and methods," *OR Spectr.*, vol. 34, no. 3, pp. 491–510, Jul. 2012.

- [23] L. A. Silman, Z. Barzily, and U. Passy, "Planning the route system for urban buses," *Comput. Oper. Res.*, vol. 1, no. 2, pp. 201–211, Aug. 1974.
- [24] D. Bertsimas, A. Delarue, and S. Martin, "Optimizing schools' start time and bus routes," *Proc. Nat. Acad. Sci. USA*, vol. 116, no. 13, pp. 5943–5948, Mar. 2019.
- [25] F. Zhao and X. Zeng, "Optimization of transit route network, vehicle headways and timetables for large-scale transit networks," *Eur. J. Oper. Res.*, vol. 186, no. 2, pp. 841–855, Apr. 2008.
- [26] J. de Dios Ortúzar and L. G. Willumsen, *Modelling Transport*. Hoboken, NJ, USA: Wiley, 2024.
- [27] N. R. Velaga, M. Beecroft, J. D. Nelson, D. Corsar, and P. Edwards, "Transport poverty meets the digital divide: Accessibility and connectivity in rural communities," *J. Transp. Geography*, vol. 21, pp. 102–112, Mar. 2012.



ZIHAO YU (Graduate Student Member, IEEE) received the B.Eng. degree in software engineering from Chiang Mai University in 2020, and the M.S. degree in information science from the Nara Institute of Science and Technology in 2023, where he is currently pursuing the Ph.D. degree in information science. His research interests include data-driven urban mobility, public transportation planning, spatial data analysis, graph theory, and image processing.



BRIAN GODWIN S. LIM (Member, IEEE) received the bachelor's and master's degrees in applied mathematics from the Ateneo de Manila University, and the Ph.D. degree in information science from Nara Institute of Science and Technology. He is currently an Assistant Professor with the Division of Information Science, Nara Institute of Science and Technology. He is affiliated with Kyoto University and Ateneo de Manila University. His research interests include mathematical modeling, graph theory, discrete algorithms, mathematical finance, econometrics, and machine learning.



RENZO ROEL P. TAN (Member, IEEE) received the Ph.D. degree in information science from Nara Institute of Science and Technology. He is currently an Assistant Professor with the Division of Information Science, Nara Institute of Science and Technology, and also affiliated with Ateneo de Manila University. His research interests include discrete algorithms, machine learning, and mathematical informatics.



HIROTAKA KAJI (Member, IEEE) received the Ph.D. degree in informatics from Kyoto University, Japan, in 2008. He was with Yamaha Motor Company Ltd., Japan, from 1998 to 2009, and has been with Toyota Motor Corporation, Japan, as a Researcher, since 2009. His research interests include machine learning, evolutionary computation, and human information processing.



KAZUSHI IKEDA (Senior Member, IEEE) received the B.E., M.E., and D.E. degrees from The University of Tokyo, in 1989, 1991, and 1994, respectively. He was with Kanazawa University, as an Assistant Professor, from 1994 to 1998. He joined Kyoto University, as an Associate Professor, in 1998. Since 2008, he has been a Professor with Nara Institute of Science and Technology. He is currently a Professor with the Division of Information Science, Nara Institute of Science and Technology. His research interests include machine learning theory, such as support vector machines and information geometry, applications to adaptive systems, and brain informatics.

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